

Table sheet

customer_id	order_id	order_date	order_value	product_category	country
C001	O1001	2024-01-15	120.50	Electronics	Bulgaria
C001	O1023	2024-03-22	89.99	Home & Garden	Bulgaria
C002	O1002	2024-02-10	45.00	Books	Greece
C003	O1003	2024-04-05	300.00	Fashion	Bulgaria
C003	O1055	2024-07-19	150.00	Fashion	Bulgaria
C004	O1004	2024-01-30	75.00	Sports Equipment	Romania

1. Calculating Customer Lifetime Value

Generating total revenue

$120.50 + 89.99 + 45.00 + 300.00 + 150.00 + 75.00 = 780.49$ (Total Revenue)

Revenue (each customer):

customer_id

C001 – $120.50 + 89.99 = 210,49$ Revenue

C002 – 45.00 Revenue

C003 – $300.00 + 150.00 = 450.00$ Revenue

C004 – 75.00 Revenue

Customer	Total Revenue
001	210.49
002	45.00
003	450.00
004	75.00

Average order value

We have customer 001 and customer 003 with more than 1 order, so:

- Customer 001 – 120.50 (order 1) + 89.99 (order 2) = $210,49 / 2 = 105,25$ (average value);
- Customer 003 – 300.00 (order 1) + 150.00 (order 2) = $450.00 / 2 = 225$ (average value)

And the average order value for the other users with only 1 order: User C002 (**45.00 average value**) and user C004 (**75.00 average value**).

Customer	Average Order Value
Customer 001	105.25
Customer 002	45.00
Customer 003	225.00
Customer 004	75.00

Purchase frequency

Customer	Total Orders
001	2
002	1
003	2
004	1

And if we find out average frequency for all customers and order, therefore:

6 orders / 4 customers = **1.5** is the total average frequency.

CUSTOMER LIFETIME VALUE

Here is the formula: (Average Order Value × Purchase Frequency) × Customer Lifespan (with an assume of 2 years)

- Customer 001: $(105.25 * 2) * 2 = 210.50 * 2 = \mathbf{421.00}$
- Customer 002: $(45.00 * 1) * 2 = \mathbf{90.00}$
- Customer 003: $(225.00 * 2) * 2 = 450 * 2 = \mathbf{900.00}$
- Customer 004: $(75 * 1) * 2 = \mathbf{150.00}$

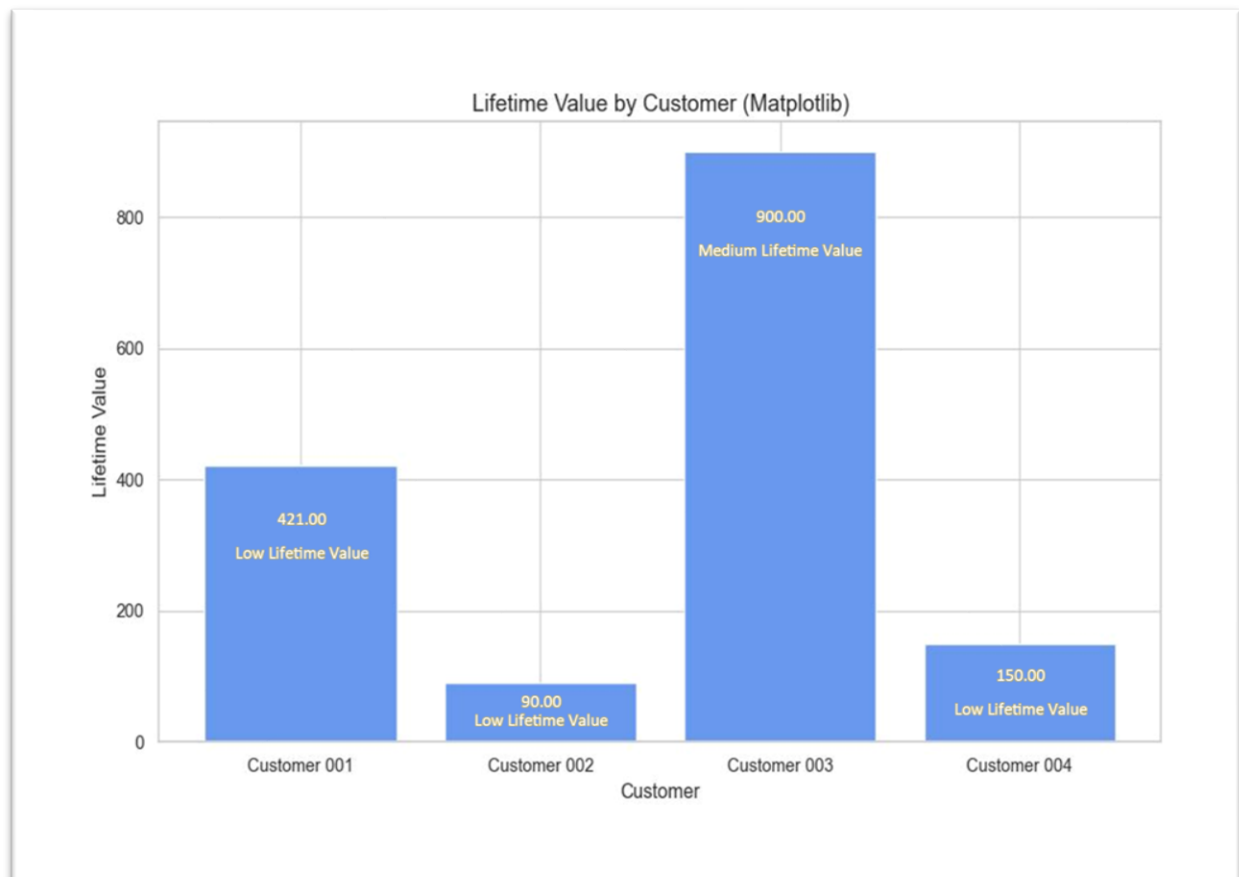
2. Segment Customers Based on CLV

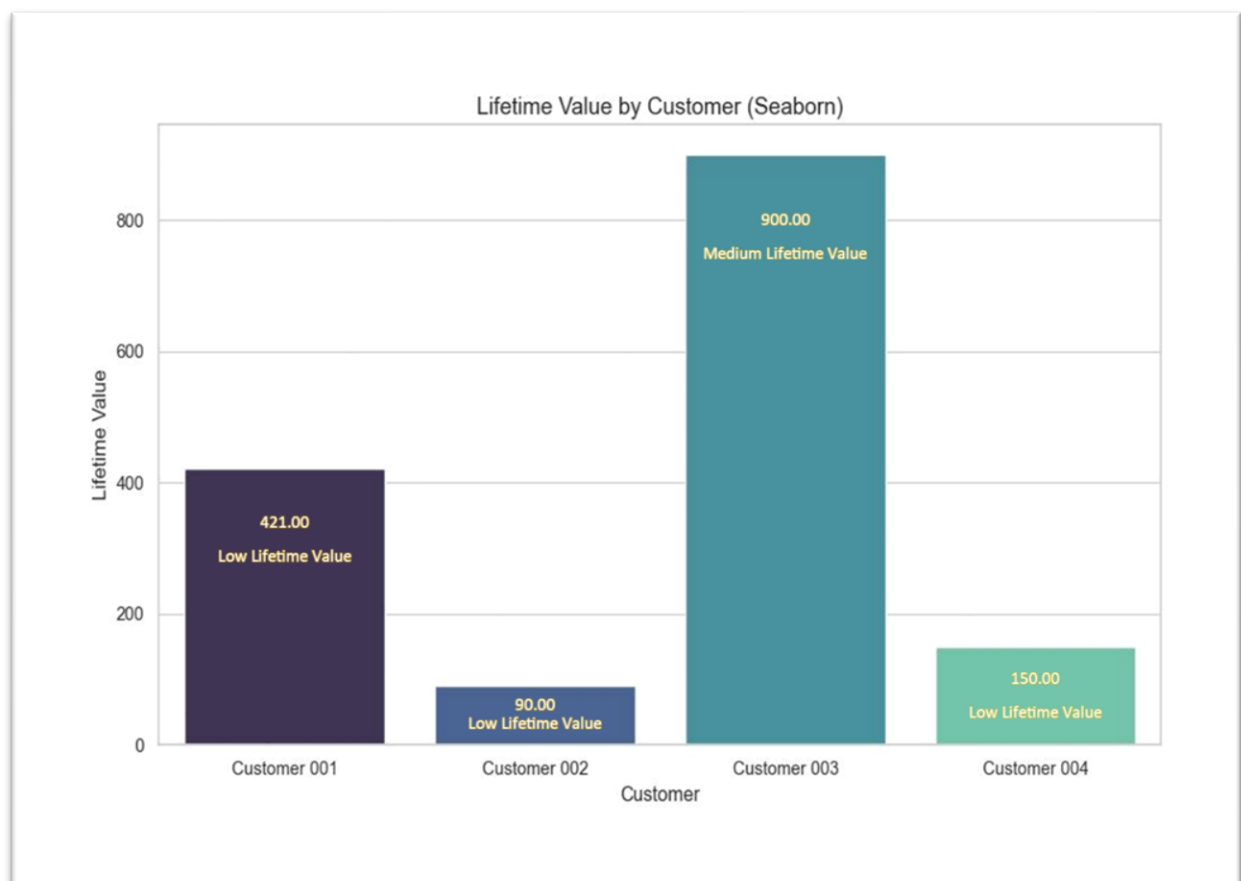
Creating a new column CLV_segment

High Value: CLV > 1000

Medium Value: CLV between 500 and 1000

Low Value: CLV < 500





Customer	Lifetime Value (Level)
Customer 001	421.00 – Low Value
Customer 002	90.00 – Low Value
Customer 003	900.00 – Medium Value
Customer 004	150.00 – Low Value

Three of four customers are with low level of Lifetime Value (under 500) and only one (customer 003) – with medium level of Lifetime Value (between 500 and 1000)

3. Time-Based Analysis

Identifying the month with the highest revenue

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So, we have the following months:

- January – 2 orders:
 - ❖ Customer C001 with order value of 120.50 (order id 1001);
 - ❖ Customer C004 with order value of 75.00 (order id 1004);
- February – 1 order:
 - ❖ Customer C002 with order value of 45.00 (order id 1002);
- March – 1 order:
 - ❖ Customer C001 with order value of 89.99 (order id 1023);
- April – 1 order:
 - ❖ Customer C003 with order value of 300.00 (order id 1003);
- July – 1 order:
 - ❖ Customer C003 with order value of 150.00 (order id 1055)

When I calculate for each month – as a result it is April with the highest order value – **300.00**

Calculating monthly revenue trends

It isn't easy to be said if there is an increasing of the prices in some of the months, because there isn't much information on this dataset, but...something interesting here:

There is been a lot of payment about **Fashion** (clothes, shoes, accessories) and those are the highest revenues. The people that spend for that – they are from Bulgaria, which means that the Bulgarian people care mostly about the outfit. About the dates of buying clothes, shoes and accessories – 300.00 money spend on 05.04.2024. That proves that they probably focus on the price discounts just before the summer season. Also on 19.07.2024 – which is possible to buy summer or autumn clothes.

A little money are spent for **Books** and that is by people from Greece. Perhaps the Greek people are investing for reading and educating.

And the Romanians spend money for **Sports equipment**. Looks like they probably care for the sport, for the health.

Detecting seasonality or anomalies

This data is just a little bit, it is hard to detect seasonality or possible anomalies in the prices. We do not know the exact quantity of bought stuff, no matter which category is.

We can say about seasonality that there is spending just before and during the summer for Fashion. People are investing for clothes, shoes, accessories. Still, many people you can see during the summer and especially when they go on a summer vacation and perhaps that is why. ***So, we have here summer seasonality in the Fashion.***

I cannot tell for sure about anomalies, we know of different prices in the money spending for Fashion, but there is no information about the quantities of bought products.

4. Retention Analysis

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Calculating the percentage of customers who made repeat purchases

We have only 4 customers and we can see that there are only 2 of them with more than one purchase – Customer C001 and Customer C003. We have 2 more with only 1 purchase.

*So, calculating by percent – we have **50%** (half) of the clients with more than 1 purchase.*

There is no information about churned customers!

Business strategies

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So, I am showing again the table visualizations.

The analysis says that there are many customers with Low Value of Lifetime, which means that they do not buy much. Well, it depends on the financial opportunities of each of them and what do they need at the moment and also the standard of the country, because that also matters much. I will suggest some strategies – what could be done for improving the revenues:

- Make a small percent of discounts if a product is being bought so often. Perhaps in that moment many people need that product;
- In some countries, people care for exact products. For example, the people from Bulgaria have a lot of interest about Fashion (buying clothes, shoes, accessories). If the place is near the sea and during the summer season, there are many Bulgarians, then you can make a bit of price reduction, to be sure that more people will buy. If there are many Greek people in that moment, you can reduce the prices about books a bit;
- It also depends on the period. People are focusing to buy outfits before and during the summer season. Then you can reduce the price in that moment.

The most important is: You must find the right moment to do that. If not, then that will lead the situation to a losing, more than winning. It is required for you to be informed, because if you are not, then you can't have good enough strategy that will take you to improvements.

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Programs and platforms used in this analysis:

- Microsoft Word (for the report);
- Microsoft Excel (for making the tables);
- Microsoft Publisher (for some corrections on the visualizations);
 - Matplotlib library (for creating the visualizations);
 - Seaborn library (for creating the visualizations);
- AI (for difficult in the understanding of some situations)